

EXPLORER

GENESYS CONVERSATIONS

> .env
> .vscode
E .funcignore
• .gitignore
• function_app.py
() host.json
() local.settings.json
E requirements.txt

function_app.py × local.settings.json

function_app.py > my_function

```
1 import datetime
2 import logging
3 import requests
4 import azure.functions as func
5 import requests
6 from requests.auth import HTTPBasicAuth
7 import pyodbc
8 import json
9 from azure.storage.blob import BlobServiceClient
10 import os
11 import pandas as pd
12 import json
13 from io import StringIO
14 import time
15 from datetime import datetime, timedelta
16 import io
17 import csv
18 from pandas import json_normalize
19 import ast
20 from sqlalchemy import create_engine
21
22 app = func.FunctionApp()
23
24 @app.function_name(name="genesysconversations")
25 @app.timer_trigger(schedule="0 14 * * *", arg_name="genesysconversations", run_on_startup=False,
26 use_monitor=False)
27 def timer_trigger(genesysconversations: func.TimerRequest) -> None:
28     custom_flag = os.environ.get("CUSTOM_FLAG")
29     custom_date = os.environ.get("CUSTOM_DATE")
30
31
32     if genesysconversations.past_due:
33         logging.info('The timer is past due!')
34
35     logging.info('Python timer trigger function executed.')
36     my_function(custom_flag, custom_date)
37
38 def my_function(custom_flag, custom_date):
39     client_id = "a39d45e8-439a-4f37-9ba9-41a902291b23"
40     client_secret = "_rfmPwaB9JOTkLW44W0gKuxdqDgC7Pk-qXdfTjfy6wQ"
41     region = 'usw2'
42
43     url = f'https://login.{region}.pure.cloud/oauth/token'
44     data = {
45         'grant_type': 'client_credentials'
46     }
47     response = requests.post(url, data=data, auth=(client_id, client_secret))
48     print(response.json())
```

> OUTLINE

> TIMELINE

0 0 0

Ln 224, Col 21



Search



Selection View Go Run Terminal Help

mtr-vdi-cp62 Ctrl+Alt+Del USB Devices Exit Fullscreen

function_app.py x local.settings.json

function_app.py > my_function

38 def my_function(custom_flag, custom_date):

47 response = requests.post(url, data=data, auth=(client_id, client_secret))

48 print(response.json())

49 print(response.json()['access_token'])

50 access_token = response.json()['access_token']

51 logging.info(access_token)

52

Run Cell | Run Below

53 # %% [markdown]

54 # **Step 3:** Get list of Conversation IDs from SQL Server

55

Run Cell | Run Above | Debug Cell

56 # %%

57

58

59 # Connection details

60 server = 'insights-westcentral-sqlserver.database.windows.net'

61 database = '1080-0GE-Production'

62 username = '0GEProductionreadonly'

63 password = 'Pr^HF#ZV'

64

65 conn_str =

66 "DRIVER={ODBC Driver 18 for SQL Server};"

67 f"SERVER={server};"

68 f"DATABASE={database};"

69 f"UID={username};"

70 f"PWD={password};"

71

72

73 try:

74 # Connect to SQL Server

75 conn = pyodbc.connect(conn_str)

76 cursor = conn.cursor()

77

78 if custom_flag == "N":

79 query = """select distinct ConversationId from AI_ConversationView_Detail

80 where cast([Interval Start] as date) = cast(getdate()-1 as date) and MediaTypeCode = 1

81 and QueueId in

82 (

83 '269249D8-F81D-4D76-9515-F94AB2B25E9D',

84 '4CEC0F08-33EB-4369-B3AB-F4698C92961A',

85 '7D8EDD53-E55C-4F47-A1FF-F03EA64E0DE2',

86 '128B5762-2DA5-4E68-AF44-E332F9B47C09',

87 '226147F7-68D1-45FF-ACDE-DDD032870FB2',

88 '71335D4E-34B0-4FE4-B2C7-C00585A54285',

89 'A9AF39E5-0BD6-41A7-B0C6-A9DC4CB85FD7',

90 '1F1FED5A-68EF-4A32-9A15-84E3646440C5',

91 'C6EA36EA-B74D-499A-AC1F-84024CD55CB4',

92

Ln 70, Col 27

EXPLORER

GENESYS CONVERSATIONS

```
> .venv
> .vscode
> .funcignore
> .gitignore
function_app.py
host.json
local.settings.json
requirements.txt
```

function_app.py X local.settings.json

function_app.py > my_function

```
38 def my_function(custom_flag, custom_date):
103     'C405D8A1-7503-4CA9-B292-1A90CB2561F9',
104     'AA4B66D3-C2BB-444B-A003-05D334344DF2'
105 );"""
106 elif custom_flag == "Y":
107     query = f"""select distinct ConversationId from AI_ConversationView_Detail
108     where cast([Interval Start] as date) = '{custom_date}' and MediaTypeCode = 1
109     and QueueId in
110     (
111         '269249D8-F81D-4D76-9515-F94AB2B25E9D',
112         '4CEC0F08-33EB-4369-B3AB-F4698C92961A',
113         '7D8EDD53-E55C-4F47-A1FF-F03EA64E0DE2',
114         '128B5762-2DA5-4E68-AF44-E332F9B47C09',
115         '226147F7-68D1-45FF-ACDE-DDD032870FB2',
116         '71335D4E-34B0-4FE4-B2C7-C00585A54285',
117         'A9AF39E5-0BD6-41A7-B0C6-A9DC4CB85FD7',
118         '1F1FED5A-68EF-4A32-9A15-84E3646440C5',
119         'C6EA36EA-B74D-499A-AC1F-84024CD55CB4',
120         '21E90526-2978-4653-BF8C-804D67F7B641',
121         '77108998-2BDE-4BF6-BB21-73674F531C50',
122         '266711DC-5643-4D57-8F2E-7347B8EFF521',
123         'C9B583C2-4043-44AB-992E-6FAD0523FFED',
124         'B5BC2EB1-0A30-4391-8C16-6E239DF08C85',
125         'BFD476A2-42D9-4596-87DA-59E18EB27ED7',
126         'CA568921-3856-4409-8446-522F01DC8598',
127         '5CC8FB4A-C912-4302-94EA-34A577A54BF7',
128         '71E9CD51-A1E0-485A-BD48-33FB49458943',
129         'C4EDB622-B9B7-414B-9BD9-30FF20C38A2F',
130         '8B04D80B-925B-4A7C-AF8C-1B6BB2DBC861',
131         'C405D8A1-7503-4CA9-B292-1A90CB2561F9',
132         'AA4B66D3-C2BB-444B-A003-05D334344DF2'
133     );"""
134
135
136     cursor.execute(query)
137
138     # Fetch results
139     conversation_ids = cursor.fetchall()
140     conversation_ids = [row for row in conversation_ids if row[0] != '']
141     logging.info('Print Conversation_ids-----.')
142     print("Print Conversation_ids-----")
143     #logging.info(conversation_ids)
144     #for row in conversation_ids:
145     #    print(row.ConversationId)
146
147     # Close connection
148     cursor.close()
149     conn.close()
```

> OUTLINE

> TIMELINE

Ln 70, Col 27 Spaces: 4

EXPLORER

GENESYS CONVERSATIONS

.venv
.vscode
.funcignore
.gitignore
function_app.py
host.json
local.settings.json
requirements.txt

function_app.py x local.settings.json

function_app.py > my_function

38 def my_function(custom_flag, custom_date):

148 cursor.close()

149 conn.close()

150

151 except Exception as e:

152 print("Error while connecting to SQL Server:", e)

153 logging.error("An error occurred: %s", e, exc_info=True)

154

155

Run Cell | Run Above | Debug Cell

%%

156

157

158

159 # Azure storage account details

160 account_url = "https://ogeprodmetranalytics.blob.core.windows.net"

161 container_name = "genesys-api"

162 blob_name = "conversations" # path inside container

163 credential = "PQ5zkVWt0N5+aCuKg+Vq209LLSCT/Hbqm1jrgFGg8T319D4P0XQ03FW1GUuItKIPYEG+yozARWS2+AsteYqimA=="

164

165 # Create Blob service client

166 blob_service_client = BlobServiceClient(account_url=account_url, credential=credential)

167

168 # Get container client

169 container_client = blob_service_client.get_container_client(container_name)

170

Run Cell | Run Above | Debug Cell

%%

171

172

173

174 if custom_flag == "N":

175 | yesterday = (datetime.now() - timedelta(days=1)).strftime("%Y-%m-%d")

176 elif custom_flag == "Y":

177 | yesterday = custom_date

178 blob_name = f"conversations/{yesterday}/.placeholder"

179

180 # Get a blob client for the new "folder"

181 blob_client = blob_service_client.get_blob_client(container=container_name, blob=blob_name)

182

183 # Upload an empty content to create the virtual folder

184 blob_client.upload_blob(b"", overwrite=True)

185 logging.info("uploading the empty file to blob")

186

Run Cell | Run Above

%% [markdown]

187

188

189

Run Cell | Run Above | Debug Cell

%%

190

191

> OUTLINE

> TIMELINE



Ln 70, Col 27 Spaces: 4 UTF-8 CRLF



Search



DELL


```
File Edit Selection View Go Run Terminal Help
mtr-vdi-cp62 Ctrl+Alt+Del USB Devices Exit Fullscreen

EXPLORER
GENESYS CONVERSATIONS
> .env
> .vscode
> .funcignore
> .gitignore
function_app.py
host.json
local.settings.json
requirements.txt

function_app.py > my_function
38 def my_function(custom_flag, custom_date):
201 def get_summary(url, headers, max_retries=3, delay=0.3):
202     for attempt in range(max_retries):
203         try:
204             response = requests.get(url, headers=headers, timeout=10)
205
206             if response.status_code == 200:
207                 return response
208             elif response.status_code == 404:
209                 # do not retry this record
210                 print(f"Attempt {attempt+1}: Skipping {url} - Not Found")
211
212             except requests.exceptions.RequestException as e:
213                 print(f"Attempt {attempt+1}: Exception for {url} - {e}")
214                 time.sleep(delay)
215             print(f"Giving up on {url} after {max_retries} attempts.")
216
217
218     headers = {
219         'Authorization': f'Bearer {access_token}',
220         'Content-Type': 'application/json'
221     }
222
223     file_path = f"conversations/{yesterday}/"
224     record_count = 1
225
226     #for conversation_id in conversation_ids:
227     for row in conversation_ids:
228         conversation_id = row.ConversationId
229         print(conversation_id)
230         url = f'https://api.usw2.pure.cloud/api/v2/conversations/{conversation_id}/summaries'
231
232         response = get_summary(url, headers)
233
234
235         if response is not None:
236             print("Response:")
237             print(response)
238             file_name = f'{conversation_id}.json'
239             blob_path = file_path+file_name
240             blob_client = blob_service_client.get_blob_client(container=container_name, blob=blob_path)
241             # Write API response into ADLS file
242             blob_client.upload_blob(response, overwrite=True)
243             print(f'Success to get data for {conversation_id}: {response.status_code}')
244             print(f'{record_count} of {len(conversation_ids)}. Saved response to {file_path+file_name}')
245         else:
246             print(f'Failed to get data for {conversation_id}')
247             print(f'{record count} of {len(conversation_ids)}. Skipped {conversation id} due to repeated failures.')
```

Ln 70, Col 27 Spaces: 4 UTF-8 CRU

EXPLORER

GENESYS CONVERSATIONS

- > .env
- > .vscode
- > .funcignore
- > .gitignore
- function_app.py
- > host.json
- > local.settings.json
- > requirements.txt

function_app.py x local.settings.json

function_app.py > my_function

```
38 def my_function(custom_flag, custom_date):
245     else:
246         print(f'Failed to get data for {conversation_id}')
247         print(f'{record_count} of {len(conversation_ids)}. Skipped {conversation_id} due to repeated failures.')
248         time.sleep(0.3) # Pause for 0.3 seconds to avoid hitting rate limits
249         record_count += 1
250
251
252
```

Run Cell | Run Above | Debug Cell

```
253 # %%
254 logging.info(f"Files in '{file_path}':")
255 for blob in container_client.list_blobs(name_starts_with=file_path):
256     print("-", blob.name)
257     logging.info(blob.name)
258
```

Run Cell | Run Above

```
259 # %% [markdown]
260 # **Step 5:** Combine json responses into a single csv file.
261
```

Run Cell | Run Above | Debug Cell

```
262 # %%
263 ### --- Combine json responses into a single csv file.
264
265
266
267
```

```
268 # List to hold data from all files
269 data = []
270
```

```
271 # Loop through all JSON files in the directory
272 #for filename in os.listdir(json_dir):
273 for blob in container_client.list_blobs(name_starts_with=file_path):
274     if blob.name.endswith('.json'):
275         print(f"Processing blob: {blob.name}")
276
277         # Get blob client
278         blob_client = container_client.get_blob_client(blob)
279
280         # Download content
281         content = blob_client.download_blob().readall()
282
283         # If JSON, load into variable
284         try:
285             file_data = json.loads(content)
286             print("JSON content loaded:", blob_path)
287         except json.JSONDecodeError:
288             print("Not a JSON file, raw content length:", len(content))
289         if isinstance(file_data, list):
```



ORER

SYS CONVERSATIONS

env

code

incignore

itignore

function_app.py

st.json

cal.settings.json

requirements.txt

function_app.py x local.settings.json

function_app.py > my_function

```
38 def my_function(custom_flag, custom_date):
39     except json.JSONDecodeError:
288         print("    Not a JSON file, raw content length:", len(content))
289     if isinstance(file_data, list):
290         data.extend(file_data)
291     else:
292         data.append(file_data)
293
294     # Use json_normalize to expand nested fields
295     df = pd.json_normalize(data)
296     #print("    normalized data ", df)
297
298
299     # Output to CSV
300     combined_file_name = f'{yesterday}.csv'
301     combined_file_path = 'conversations_combined/'
302     combined_blob_path = combined_file_path+combined_file_name
303
304     csv_buffer = StringIO()
305     df.to_csv(csv_buffer, index=False)
306
307     blob_client = blob_service_client.get_blob_client(container=container_name, blob=combined_blob_path)
308     # Write API response into ADLS file
309     blob_client.upload_blob(csv_buffer.getvalue(), overwrite=True)
310     print(f'Success to write data for {combined_file_name}')
311
312
313     Run Cell | Run Above
314     # %% [markdown]
315     # **Step 6:** Expand all json responses and double-quote each csv column.
316
317     Run Cell | Run Above | Debug Cell
318     # %%
319     ### ----- Expand all json responses and double-quote each csv column.
320
321     print(f"📁 Files in '{combined_blob_path}':")
322     for blob in container_client.list_blobs(name_starts_with=combined_blob_path):
323         print(" -", blob.name)
324
325     # Get blob client
326     blob_client = container_client.get_blob_client(blob)
327
328     # Download content
329     download_stream = blob_client.download_blob()
330     csv_bytes = download_stream.readall()
331     csv_str = csv_bytes.decode("utf-8")
332     df = pd.read_csv(StringIO(csv_str))
```

LINE

LINE

EXPLORER

GENESYS CONVERSATIONS

- venv
- vscode
- funcignore
- .gitignore
- function_app.py
- host.json
- local.settings.json
- requirements.txt

function_app.py x local.settings.json

function_app.py > my_function

```
330 def my_function(custom_flag, custom_date):
331     csv_str = csv_bytes.decode("utf-8")
332     df = pd.read_csv(StringIO(csv_str))
333
334
335     # Parse the sessionSummaries column (string to list of dicts)
336     def parse_session_summaries(val):
337         try:
338             return ast.literal_eval(val)
339         except Exception:
340             return []
341
342     # Expand the first session summary in each row
343     expanded = df['sessionSummaries'].apply(parse_session_summaries).apply(lambda x: x[0] if x else {})
344
345     # Create a DataFrame from the expanded session summaries
346     expanded_df = pd.json_normalize(expanded)
347
348     # Combine with the other columns (excluding the original sessionSummaries)
349     result = pd.concat([expanded_df, df.drop(columns=['sessionSummaries'])], axis=1)
350
351     # remove data from the column editedSummary.text to avoid exposing any PII
352     result['editedSummary.text'] = ''
353
354     # drop any row with an error type of 'ConversationTooShort'
355     try:
356         result = result[result['errorType'] != 'ConversationTooShort']
357     except Exception as e:
358         print(e)
359         logging.error("An error occurred: %s", e, exc_info=True)
360
361
362     # remove line breaks for the text column
363     result['text'] = result['text'].str.replace('\n', ' ')
364
365     # remove quote characters
366     result['text'] = result['text'].str.replace('"', '')
367
368
369     # Output to CSV
370     expanded_file_name = f'{yesterday}.csv'
371     expanded_file_path = 'conversations_expanded/'
372     expanded_blob_path = expanded_file_path+expanded_file_name
373
374     csv_buffer = StringIO()
375     result.to_csv(csv_buffer, index=False)
376
377     blob_client = blob_service_client.get_blob_client(container=container_name, blob=expanded_blob_path)
```

OUTLINE

TIMELINE

A D

Ln 70, Col 27 Spaces: 4 UTF-8 CRLF



Search



EXPLORER

GENESYS CONVERSATIONS

- .env
- .vscode
- .funcignore
- .gitignore
- function_app.py
- host.json
- local.settings.json
- requirements.txt

function_app.py X local.settings.json

function_app.py > my_function

```
375 def my_function(custom_flag, custom_date):
376     result.to_csv(csv_buffer, index=False)
377
378     blob_client = blob_service_client.get_blob_client(container=container_name, blob=expanded_blob_path)
379     # Write API response into ADLS file
380     blob_client.upload_blob(csv_buffer.getvalue(), overwrite=True)
381     print(f'Success to write data for {expanded_file_name}')
382     logging.info(f'Step 6: Success to write data for {expanded_file_name}')
383
384
385
386
387 Run Cell | Run Above
388 # [markdown]
389 # **Step 7:** Get the call attributes. Grab the list of conversation ids from the expanded_output file.
390
391 Run Cell | Run Above | Debug Cell
392 # [markdown]
393 ### ---- Get the call attributes. Grab the list of conversation ids from the expanded_output file.
394 def get_convo_detail(detail_url, headers, max_retries=3, delay=0.3):
395     for attempt in range(max_retries):
396         try:
397             resp = requests.get(detail_url, headers=headers, timeout=10)
398             resp.raise_for_status()
399             return resp.json()
400         except requests.exceptions.RequestException as e:
401             print(f'Attempt {attempt+1} failed: {e}')
402             time.sleep(delay)
403     print(f'Failed to get data from {detail_url} after {max_retries} attempts.')
404     return None
405
406 print(f'Files in {expanded_blob_path}:')
407 logging.info(f'Files in {expanded_blob_path}:')
408 for blob in container_client.list_blobs(name_starts_with=expanded_blob_path):
409     print("-", blob.name)
410
411 # Get blob client
412 blob_client = container_client.get_blob_client(blob)
413
414 # Download content
415 download_stream = blob_client.download_blob()
416 csv_bytes = download_stream.readall()
417 csv_str = csv_bytes.decode("utf-8")
418 df = pd.read_csv(StringIO(csv_str))
419
420 #create a list of conversation ids to use
```



EXPLORER

NESYSCONVERSATIONS

.venv
vscode
funcignore
.gitignore
function_app.py
host.json
local.settings.json
requirements.txt

function_app.py X local.settings.json

function_app.py > my_function

```
38 def my_function(custom_flag, custom_date):
418
419
420     #create a list of conversation ids to use
421     convo_ids = df['conversation.id'].tolist()
422     print(convo_ids)
423
424
425     # List of keys to extract for customer attributes
426     keys_to_extract = [
427         "IVRInfo.SERVICE_ADDRESS",
428         "IVRInfo.LAST_PAY_AMOUNT_IN2",
429         "IVRInfo.TOTAL_AMOUNT",
430         "IVRInfo.LAST_PAY_AMOUNT",
431         "IVRInfo.LAST_PAY_DATE",
432         "IVRInfo.NEXT_MAIL_DATE",
433         "SmartHours",
434         "DNIS",
435         "AMB",
436         "IVRInfo.ZIP_CODE",
437         "IVRInfo.NEXT_METER_READ",
438         "IVRPath",
439         "IVRInfo.DUE_DATE",
440         "IVRInfo.AMOUNT_TO_RECONNECT",
441         "Priority",
442         "IP",
443         "IVRInfo.PAST_DUE",
444         "Eligibility.CUST_PROGRAMS",
445         "IVRInfo.STATE",
446         "InstallmentPlan.OPT_OptionsData",
447         "IVRInfo.BUSINESS_FLAG",
448         "ANICConfirmed",
449         "IVRInfo.HOUSE_NUM",
450         "ANI",
451         "ScreenPopName",
452         "InstallmentPlan.OPT_ERROR_MESSAGE",
453         "Authenticated",
454         "Paperless",
455         "IVRInfo.PAST_DUE_OVER10",
456         "HEEP",
457         "InstallmentPlan.FINALED",
458         "IVRInfo.RECORD_FOUND",
459         "InstallmentPlan.OPT_CURRENT_AMOUNT_DUE",
460         "Reason",
461         "InstallmentPlan.OPT_PAST_DUE",
462         "IVRInfo.CONTRACT_ACCOUNT",
463         "IVRInfo.DEPOSIT_DUE",
464         "Zero Out Count",
465         "language"
```

OUTLINE

MELINE



function_app.py local.settings.json

function_app.py > my_function

```

38 def my_function(custom_flag, custom_date):
484     "IVInfo.BUSINESS_PARTNER",
485     "Last4Check.RETURN_CODE"
486 ]
487
488
489
490
491 # Set up the headers, including the authorization token
492 headers = {
493     'Authorization': f'Bearer {access_token}',
494     'Content-Type': 'application/json'
495 }
496
497 # place the ids into batches of 50
498 batches = [convo_ids[i:i + 50] for i in range(0, len(convo_ids), 50)]
499
500 # Set to keep track of unique conversation IDs
501 unique_conversations = set()
502
503 output = StringIO()
504 output_obj = []
505 header = ["conversation_id"] + keys_to_extract
506 for batch in batches:
507     #pause for 1 second between batches
508     time.sleep(0.3)
509     # print the number of the batch being processed
510     print(f"Processing batch {batches.index(batch) + 1} of {len(batches)} : {batch}")
511
512 #loop through the convo_ids
513 for conversation_id in batch:
514     print(conversation_id)
515     # sleep for 1/2 seconds to avoid hitting API rate limits
516     time.sleep(0.3)
517     # ---- Get full conversation detail ----
518     detail_url = f"https://api.{region}.pure.cloud/api/v2/conversations/{conversation_id}"
519     #detail_resp = requests.get(detail_url, headers=headers)
520     #convo_detail = detail_resp.json()
521     convo_detail = get_convo_detail(detail_url, headers)
522
523     if convo_detail is None:
524         continue # skip this conversation_id
525
526     customer_attrs = {}
527     for part in convo_detail.get("participants", []):
528         purpose = part.get("purpose")
529         if purpose == "customer":
530             customer_attrs = part.get("attributes", {})
531             # Extract the needed info

```



function_app.py X local.settings.json

function_app.py > my_function

```
38 def my_function(custom_flag, custom_date):
529     if purpose == "customer":
530         customer_attrs = part.get("attributes", {})
531         # Extracting the specified keys
532         # if a key is not present in the customer_attrs, it will be skipped
533         extracted_data = {key: customer_attrs[key] for key in keys_to_extract if key in customer_attrs}
534
535         row = [conversation_id]
536         row_temp = []
537         for col_name in header:
538             val = str(extracted_data.get(col_name, conversation_id if col_name == "conversation_id" else "")).replace("\n", " ").replace("\r", " ")
539             row_temp.append(val)
540
541         # if the length of the line is greater than 250 characters, take it, otherwise skip it
542         if len(",".join(row)) > 250:
543             row.append(extracted_data.get(key, ""))
544
545         for key in keys_to_extract:
546             row.append(extracted_data.get(key, ""))
547
548         # Write header if it's the first row
549         if len(unique_conversations) == 0:
550             writer = csv.writer(output, quoting=csv.QUOTE_ALL)
551             writer.writerow(header)
552             output_obj.append(header)
553         else:
554             writer = csv.writer(output, quoting=csv.QUOTE_ALL)
555             # Write the row as CSV with all columns quoted
556             writer.writerow([str(item).replace('\n', ' ').replace('\r', ' ') for item in row])
557             # display the row
558             #print(row)
559             output_obj.append(row)
560             #@@output_obj is a list object
561             # Add the conversation ID to the set of unique conversations
562             unique_conversations.add(conversation_id)
563
564
565
566
567     # remove any blank rows for the csv file
568     lines = [item for item in output_obj if item not in ("", None, [])]
569     lines = [item for item in lines if any(cell.strip() for cell in row)]
570     # if the length is less than 250 delete the row
571     #lines = [item for item in lines if len(item) >= 250]
572
573     lines = [
574         item for item in lines
575         if any(col not in (None, '') for col in item[1:]) # check columns after first
```




function_app.py X {} local.settings.json

function_app.py > my_function

```
38 def my_function(custom_flag, custom_date):
571     #lines = [item for item in lines if len(item) >= 250]
572
573     lines = [
574         item for item in lines
575         if any(col not in (None, '') for col in item[1:]) # check columns after first
576     ]
577
578     # test = []
579     # for item in lines:
580     #     for col in item[1:]
581
582     print(lines)
583
584
585
586     # Remove any duplicate rows
587     seen = set()
588     unique_list = []
589     for row in lines:
590         row_tuple = tuple(row) # convert list to tuple (hashable)
591         if row_tuple not in seen:
592             seen.add(row_tuple)
593             unique_list.append(row)
594
595
596
597
598     Run Cell | Run Above | Debug Cell
599     # %%
600
601     # Output to CSV
602     attributes_file_name = f'{yesterday}.csv'
603     attributes_file_path = 'conversations_attributes/'
604     attributes_blob_path = attributes_file_path+attributes_file_name
605
606     output = io.StringIO()
607     writer = csv.writer(output, quoting=csv.QUOTE_ALL)
608     for row in unique_list:
609         writer.writerow(row)
610
611
612     blob_client = blob_service_client.get_blob_client(container=container_name, blob=attributes_blob_path)
613     blob_client.upload_blob(output.getvalue(), overwrite=True)
614     #print(unique_list)
615
616
```

function_app.py x local.settings.json

VERSIONS

app.py

ngs.json

ents.txt

```
function_app.py > my_function
38 def my_function(custom_flag, custom_date):
612 blob_client = blob_service_client.get_blob_client(container=container_name, blob=attributes_blob_path)
613 blob_client.upload_blob(output.getvalue(), overwrite=True)
614 #print(unique_list)
615
616
617 print(f'Success to write data for {expanded_file_name}')
618 logging.info(f'Success to write data for {expanded_file_name}')
619
Run Cell | Run Above | Debug Cell
620 # %%
621
622
623 #Output to SQL
624
625 # Connection details test
626 #server = 'sql-test-meteranalytics.database.windows.net'
627 #database = 'SQL-test-meteranalytics-db'
628 #username = 'sqladminuser'
629 #password = 'K3QAY8YoUJVx66'
630 #driver = 'ODBC Driver 18 for SQL Server'
631
632 # Connection details prod
633 server = 'oge-prod-meteranalytics.database.windows.net'
634 database = 'OGEMeterAnalyticsProd'
635 username = 'sqladminuser'
636 password = 'YaAKEjYLw8SpZ9'
637 driver = 'ODBC Driver 18 for SQL Server'
638
639 conn_str = (
640     "DRIVER={ODBC Driver 18 for SQL Server};"
641     f"SERVER={server};"
642     f"DATABASE={database};"
643     f"UID={username};"
644     f"PWD={password};"
645 )
646
647 try:
648     # Connect to SQL Server
649     conn = pyodbc.connect(conn_str)
650     cursor = conn.cursor()
651
652     # delete query
653     truncate_query = "Truncate table genesys.CONVERSATIONS_ATTRIBUTES_STG"
654     cursor.execute(truncate_query)
655     conn.commit()
656     cursor.close()
657     conn.close()
658     print("truncated STG table")
```



function_app.py x local.settings.json

function_app.py > my_function

```
38 def my_function(custom_flag, custom_date):
656     cursor.close()
657     conn.close()
658     print("truncated STG table")
659     logging.info("truncated STG table")
660
661 except Exception as e:
662     print("Error while delete in SQL Server:", e)
663     logging.error("An error occurred: %s", e, exc_info=True)
664
665 try:
666     #print(output.getvalue())
667     output.seek(0)
668     df = pd.read_csv(output)
669     df.insert(0, "Conversation_Date", yesterday)
670     #display(df)
671     # Connect with SQLAlchemy
672
673
674     # Create connection string
675     conn_str = (
676         f"mssql+pyodbc://{username}:{password}@{server}:1433/{database}"
677         f"?driver={driver.replace(' ', '+')}&Encrypt=yes&TrustServerCertificate=no&Connection Timeout=30"
678     )
679
680     # Create SQLAlchemy engine
681     engine = create_engine(conn_str)
682
683     # Insert into table (if the table doesn't exist, it will be created)
684     df.to_sql('CONVERSATIONS_ATTRIBUTES_STG', schema="genesys", con=engine, if_exists='append', index=False)
685     print("data insterted in sql")
686     logging.info("data inserted in sql")
687
688 except Exception as e:
689     print("Error while inserting in SQL Server:", e)
690     logging.error("An error occurred: %s", e, exc_info=True)
691
692 try:
693     # Connect to SQL Server
694     conn_str = (
695         "DRIVER={ODBC Driver 18 for SQL Server};"
696         f"SERVER={server};"
697         f"DATABASE={database};"
698         f"UID={username};"
699         f"PWD={password};"
700     )
701     conn = pyodbc.connect(conn_str)
702     cursor = conn.cursor()
703
```